

Malothu Likhith

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Professional Experience

AI/ML Intern | APTRANSCO (Andhra Pradesh Transmission Corporation)

May 2026 – Present

- 800+ internal PDFs across 19 collections required manual document lookup by employees; built a **Hybrid Agentic RAG system** using a multi-agent **LangGraph workflow** (Router → Retrieval → Verification → Response) with hybrid vector + BM25 search, CrossEncoder reranking, and hallucination prevention on **AWS Bedrock** — now actively used by APTRANSCO staff for instant document retrieval, achieving **Faithfulness 0.87** and **Hallucination Rate 0.13**.
- Baseline v1 produced unreliable outputs; ran **5 iterative optimization rounds** on chunking strategy, reranking thresholds, and prompt engineering — improved **Faithfulness by +39%** and cut **Hallucination by -61%**.
- Deployed full stack (**FastAPI + React**) via **Docker on AWS EC2**; committed pre-built Qdrant vector embeddings to Git enabling **zero re-ingestion, instant cold-start deployment** for the production environment.

Web Developer Intern | Basil Infotech [\[Cert\]](#)

May 2025 – Aug 2025

- Hospital operations lacked a unified digital system; built a **scalable MERN-stack management platform** supporting **200+ records** with real-time ambulance tracking, appointment scheduling, and **MongoDB aggregation pipeline**-powered analytics dashboards.
- Designed **RESTful APIs** improving backend modularity; optimized MongoDB queries for real-time analytics and improved cross-device responsiveness using **React.js**.

Projects

RAG-Based Student Q&A Chatbot [\[GitHub\]](#)

Python | FastAPI | LangChain | ChromaDB | Hugging Face

- 680+ college documents across all years were only accessible by manual search; built a **retrieval-augmented generation chatbot** using **ChromaDB** as the vector store with semantic chunking, multi-turn conversational memory, and response caching — achieved **~85% answer accuracy** and cut retrieval latency by **~30%**.
- System was tightly coupled to a single LLM provider; redesigned into a **modular architecture** decoupling retrieval, memory, and generation layers — enabling plug-and-play integration with multiple LLM providers and improving retrieval precision by **~25%**.

SolarSense — Solar Panel Efficiency Detection System

Python | FastAPI | React | TypeScript | XGBoost | Docker

[\[GitHub\]](#) [\[Live\]](#)

- No real-time system existed to monitor college solar panel health; built a **full-stack ML application** over **2–3 years of real IV-curve data** from **7 college panels** (Mono PERC, Polycrystalline, Bifacial, TOPCon) that returns power output, fault status, remaining useful life, and a **10-year degradation forecast** with color-coded end-of-life zones.
- Trained **3 models**: XGBoost Pmax regressor ($R^2 \approx 0.87$, RMSE reduced **~18%** via GridSearchCV, 16 engineered features including solar angle proxy and 7-day rolling stats), XGBoost remaining life estimator (linear degradation on monthly Pmax trends, 80% rated capacity as end-of-life threshold), and **two-layer fault detector** — Isolation Forest (5% contamination) + rule-based labels (LOW_FF, LOW_PMAX, HIGH_SERIES_R, SHADING_OR_SHUNT) flagging **~5% of readings** as anomalies.
- Built **batch CSV/Excel upload pipeline** returning per-row predictions, degradation rate, projected power decay chart, and fault distribution — containerized with **multi-stage Docker** (Next.js static export served from FastAPI) and deployed to **Render**.

Education

B.Tech in Electrical Engineering | IIT Bhilai

2023 – 2027

Minor in Computer Science | IIT Mandi [\[Cert\]](#)

2025 – 2026

Technical Skills

Languages: Python, SQL

Generative AI & LLMs: RAG, Agentic RAG, LangChain, LangGraph, Multi-Agent Systems, Prompt Engineering, RAGAS Evaluation, Hallucination Mitigation, Embeddings, Transformers, Generative AI

ML & AI: PyTorch, Scikit-learn, XGBoost, Regression, Classification, Neural Networks, Anomaly Detection, Cross-Validation, Hyperparameter Tuning

Libraries: NumPy, Pandas, Matplotlib, Hugging Face, LangChain, PyTorch, Scikit-learn

Tools & Infra: AWS Bedrock, AWS EC2, Docker, Qdrant, ChromaDB, FastAPI, MongoDB, Git, Jupyter, VS Code, Postman

Certifications

Adobe Hackathon Round 1 [\[Cert\]](#) · **Flipkart GRiD 6.0 Level 1** [\[Cert\]](#)

Positions of Responsibility

Convener — Prayatna 8.0, IIT Bhilai Sports Fest · **Team Member** — MERAZ Cultural Fest, IIT Bhilai · **Languages:** English (Fluent), Hindi (Fluent), Telugu (Native)